BLAQUE BAUX · RESEARCH COMPENDIUM

The complete method

The full body of work behind a single-strategy platform: every approach explored, validated, or rejected — and the discipline that decided. Twenty-plus strategies, one honest scorecard, and the reason only a few of them ship.

by **Carter Warrens** · 2026 · reference implementation in Julia · figures are historical backtests / paper results on ETF and equity data

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THE PROGRAM IN ONE PARAGRAPH

Blaque Baux began as an alpha-prediction system and became a risk-premium harvester, because **rigorous out-of-sample testing found no exploitable predictive edge** at the horizons and instruments we can access — a result we reproduced a dozen ways (cross-sectional factors, Bayesian filtering, 15-minute signals, cross-industry lead-lag, pairs arbitrage: all coin flips). What *survives* is **risk structure** — diversification and trend premia, harvested by a governed daily process. Around that core we ran a deep second program on **convexity and tail protection**, on **correlation structure**, and on **aggression and leverage**, each ending in a durable, often counter-intuitive law. This paper is the map of all of it: what shipped, what is in live testing, what was kept as research, and — just as prominently — what was tested and rejected, with the reason. The honesty is the point: a strategy is only as trustworthy as the ideas it was willing to kill.

1 The founding finding

On 60-day 15-minute data (~1,547 bars × 60 S&P names), two independent alpha models scored **50.9%** and **50.7%** directional accuracy — coin flips, information coefficient ≈ 0. We then reproduced the null across blue-chip and mid-cap universes, cross-industry lead-lag, and pairs mean-reversion. The conclusion was not "try harder"; it was **structural**: at retail-accessible horizons and costs, the market has already priced what's knowable. So the platform stopped forecasting returns and started harvesting the premia paid for *bearing risk* — diversification, trend, term/asset premia — where an out-of-sample edge genuinely survives.

2 The scorecard

Every strategy and idea explored, with its verdict and status. Green = live or graduating; amber = kept as research; red = tested and rejected. This is the whole program at a glance.

STRATEGY / IDEA	CATEGORY	FINDING	STATUS
LIVE & PRODUCTION			
The Spine (inverse-vol/ERC base + 12-mo TSMOM trend + vol-target + regime brake)	Risk-premium harvest	Sharpe ≈ 0.97, maxDD ≈ −11%, 2022 −1.4%; trend sleeve carries crises. Forecasts nothing.	LIVE
DBA (agriculture) 6th sleeve	Diversification	Only genuinely-uncorrelated add tested (0.44 vs DBC); Sharpe 0.94→1.04.	LIVE
GRADUATED — IN LIVE PAPER A/B (3 ISOLATED ACCOUNTS)			
Multi-horizon trend (3/6/12-mo agreement)	Funded convexity	Strict improvement: Sharpe 1.04→1.16, drawdown −8.9→−7.0%, more convex, no cost.	A/B
Split-universe spine (base vs trend on different universes)	Breadth	Trend on a broad shortable CTA set → Sharpe ≈ 1.12, lower vol; substitutes with multi-horizon.	A/B
REVIEWED, KEPT AS RESEARCH (NOT DEPLOYED)			
Taleb barbell / the "curveball" (90/10 and reversed 10/90 convex sliver)	Convexity	A dark horse: bleeds, ruin if compounded, but +237% in COVID on its 10%. Hold, don't time.	RESEARCH
Inverse carry (short high-carry = short the premium)	Tail hedge	Positive-skew hedge that catches the rate/credit crisis (2022) VIXY misses.	RESEARCH
Diversified tail hedge (across crisis regimes)	Convexity	Deflation + inflation hedges are anti-correlated → a sliver positive in *every* crisis.	RESEARCH
Gamma-ARMA regime framework (Bayesian regime over ARMA/GARCH space)	Regime / risk	Sophisticated, specified, *unvalidated* (backtest never run). Path-A funded research.	RESEARCH
Optimizer library (Markowitz, Black-Litterman, CVaR/CDaR, HRP, Michaud, cost-aware)	Infrastructure	General-purpose portfolio-optimization stack + REST service; the spine is one strategy on it.	TOOL
TESTED & REJECTED — RETURN PREDICTION (ALPHA)			
3-factor cross-sectional & Bayesian (Kalman/NIG/James-Stein/ADVI) alpha	Alpha	50.9% / 50.7% accuracy, IC ≈ 0 — coin flips. Archived.	REJECTED
Blue-chip & mid/small-cap directional prediction	Alpha	Blue-chip a coin flip; mid-cap only short-horizon reversal (not tradeable at cost).	REJECTED

STRATEGY / IDEA	CATEGORY	FINDING	STATUS
15-minute intraday alpha	Alpha	Noise (50.9%); intraday cadence adds nothing — diversification is the shock absorber.	**REJECTED**
Lead-lag / earnings read-through (NVDA→semis, banks, tech)	Alpha	Priced same-day (corr 0.7–0.9), next-day ≈ 0; in tech it *reverses* (chasing loses).	**REJECTED**
Pairs / relative-value mean-reversion (stat-arb)	Alpha	Diversified book ≈ 0.15 Sharpe (noise); many "pairs" drift apart permanently.	**REJECTED**
TESTED & REJECTED — CONVEXITY, CARRY, LEVERAGE			
Vol-aware VIXY tail overlay on the Spine	Paid convexity	Hurts — the Spine's trend is already a crisis hedge; VIXY carry is a drawdown machine.	**REJECTED**
Convex response function (size by trend strength²)	Convexity	Backfires — levers calm uptrends, flattens skew. Convexity must be in the payoff.	**REJECTED**
Carry sleeve as a third premium	Risk premium	Genuinely uncorrelated (−0.02) but negative-expectancy + negative skew → lowers Sharpe.	**REJECTED**
Carry as the barbell's earning sleeve	Convexity	Crashes when the convex sleeve pays — eats the payoff. Safe sleeve must be inert.	**REJECTED**
Leverage to double-digit returns	Aggression	Net-negative at retail financing; Kelly ≈ 1×; higher leverage = ruin, not riches.	**REJECTED**

3 What's live, and what's graduating

Production: the Spine — a two-sleeve book (inverse-vol/ERC base + 12-month time-series-momentum trend), each volatility-targeted on its own realized P&L, blended 50/50 and scaled by a drawdown regime brake, rebalanced once daily and routed through a governed execution layer. Net of costs it delivers a Sharpe near 1.0 at ~11% maximum drawdown, with the trend sleeve making money in 2022 while a long book bled. It now trades six assets (SPY / IEF / TLT / GLD / DBC / DBA).

In live A/B (three isolated paper accounts): single-sign (the control), split-universe, and single-multi-horizon — so the two candidate upgrades are measured head-to-head against production on real fills, each with its own state, ledger, and safety files. Nothing graduates to funded capital until it clears out-of-sample / purged cross-validation.

4 The research themes

Return prediction — the null result, five ways

Every route to forecasting returns landed in the same place: cross-sectional factors, Bayesian filtering, blue-chip and mid-cap direction, 15-minute signals, cross-industry lead-lag, and pairs arbitrage. The consistency *is* the evidence — the market prices what's knowable, instantly.

Convexity & tail hedging — free vs paid

The deepest thread. Convexity comes in two forms: **free** (trend/momentum, funded by the momentum premium — protects sustained crises, improves return) and **paid** (long-vol/short-carry, funded by decay — protects sharp crashes, but bleeds and can't be timed). A paid overlay *hurts* the Spine (it already owns free convexity); making the trend sleeve multi-horizon is the win. The best standalone hedge is **diversified across crisis regimes** — deflation (VIXY, short-credit) and inflation (short-duration, long-commodities) are anti-correlated, so a small sliver has no blind spot.

Correlation structure — real, but priced

Industries, supply chains, and firms are strongly correlated (sectors ~0.62, banks ~0.87) — but the read-through is *instantaneous*, so it's a risk tool, not an alpha source. And correlation $\propto$ **shared forced exposure**: prudential regulation (banks, utilities) or a common input (airlines: fuel) homogenizes firms into one factor, while product regulation with idiosyncratic outcomes (pharma, biotech under the FDA) leaves them uncorrelated despite heavy oversight.

Aggression & leverage — the growth-vs-ruin frontier

"Most aggressive" has a principled ceiling (Kelly) and a brutal catch: aggression multiplies edge, and ours is modest. At retail financing, leverage *lowers* growth; a max-aggressive convex bet has a median terminal wealth below the stake. Rational aggression is fractional Kelly on real edge — or a small barbell sleeve — never all-in.

The archived alpha track

The original Bayesian alpha engine, a Monte-Carlo risk simulator, a Sharpe-on-historical-means "Oracle," and the Gamma-ARMA regime framework are preserved but archived — the mathematics is sound, the measured edge was not there. Sophisticated machinery *packages and sizes* a signal; it does not *create* one.

5 The durable laws

The findings that survive the small-sample caveats — directional and structural, the real output of the program.

I. At tradeable horizons there is no exploitable return-prediction edge — so harvest premia, don't forecast.

II. Convexity is *free* (trend → sustained crises) or *paid* (long-vol → sharp crises); no instrument gives cheap, timed, crash-day protection.

III. Timing the tail removes the tail — every rule that trims a hedge's bleed also removes its payoff.

IV. Diversification is necessary but not sufficient — an uncorrelated stream only helps if it is positive-expectancy.

V. A hedge's safe sleeve must be inert in a crisis; anything with hidden crisis-beta is not safe.

VI. There are two crisis regimes (deflation, inflation); they are anti-correlated and need opposite hedges.

VII. Aggression multiplies edge — with none, it multiplies only ruin (Kelly ≈ 1× here).

VIII. Correlation is real and priced instantly — a risk tool, not an alpha source.

IX. Correlation ∝ shared forced exposure (prudential regulation or a common input); idiosyncratic outcomes decorrelate.

X. The discipline — validate out-of-sample, then deploy — is the actual product; a strategy is only as trustworthy as the ideas it killed.

6 Governance — the discipline is the product

Every order passes one governed path enforcing hard, tested invariants: idempotency, per-pool budget and daily-loss limits, data-staleness, position reconciliation, a kill switch, and full fill lineage — beneath a pre-trade safety gate (drawdown/loss halts, gross-leverage and per-name caps, account and data checks). No language model is anywhere near the order path; the strategy is reproducible code. The autonomous daily job runs deterministically pre-open with morning catch-up fires, and reports its book and P&L to a comparison log. Nothing reaches capital without clearing the same out-of-sample bar the Spine did — which is why most of the scorecard is red, and that is a feature.

7 Honest caveats

Backtests are historical / paper results on ETF and equity proxies, net of modeled costs; they are not indicative of future results. Tail and earnings findings are *especially* sample-dependent — the window holds a single sharp crash (COVID) and a single inflation bear (2022). Several "positive" research results (inverse carry, short-duration hedges) lean on that 2022 window and should be read as *shapes*, not promises. The validated core is single-digit-CAGR, ~1.0-Sharpe money; there is no double-digit-return-at-low-risk configuration, and this program says so plainly.

Companion documents — validated methods & math: **docs/whitepaper.html**, **docs/FINANCIAL_METHODS.md**. Research notes: **docs/research_thread_summary.pdf**, **docs/barbell_note.pdf**, **docs/research_frontier.html**. Decisions: **docs/CANONICAL_ARCHITECTURE.md**. Reproducible sketches: **scripts/research/** (11 scripts). Reference implementation (Julia): the Blaque Baux repository.